



The no way transfer part 2

The difference between suspended mass geometric and elastic load transfer, and the influence of kinematics

BY CLAUDE ROUELLE

In this follow on from last month's article, we will explain the difference between the suspended mass geometric and elastic load transfers, and why kinematics has a considerable influence on car behaviour at the corner entrance.

In part one of this article, we divided the total lateral load transfer in a non-suspended mass load transfer and a suspended mass load transfer. A reminder is shown in **Figure 1**, as well as in part (a) of **Figure 2**.

We will now split the suspended mass lateral load transfer in a geometric and an elastic load transfer as indicated in part (b) of **Figure 2**.

Based on the principle that a force applied at one point of a body is equal to the same force applied at another point of the same body force and a torque, as shown in **Figure 3**, we will now replace the lateral force applied at the suspended mass c of g (in blue in **Figure 4**) in an equal force applied at the kinematic roll centre (in red in **Figure 4**) and a suspended mass roll moment (in yellow in **Figure 4**).

There are many considerations to be made here:

1. The roll moment that creates the suspended mass elastic load transfer is a function of the vertical distance between the suspended mass c of g and the kinematic roll centre. The higher the roll centre, the less roll moment and elastic load transfer, but the more geometric load transfer.
2. The figures shown are a simplified 2D view (here, the front) in steady state. We will soon make considerations on transient load transfers and, later, in 3D (four wheels, the whole car) load transfers.
3. As I insisted in a previous article, there are flaws in this simplified explanation. The suspended mass does *not* rotate about the kinematic roll centre. You will notice that, so far, we have only looked at the effect the lateral acceleration has on the variation of the tyres' vertical load.

Fig 1: Simplified 2D suspended mass and non-suspended mass load transfer equations

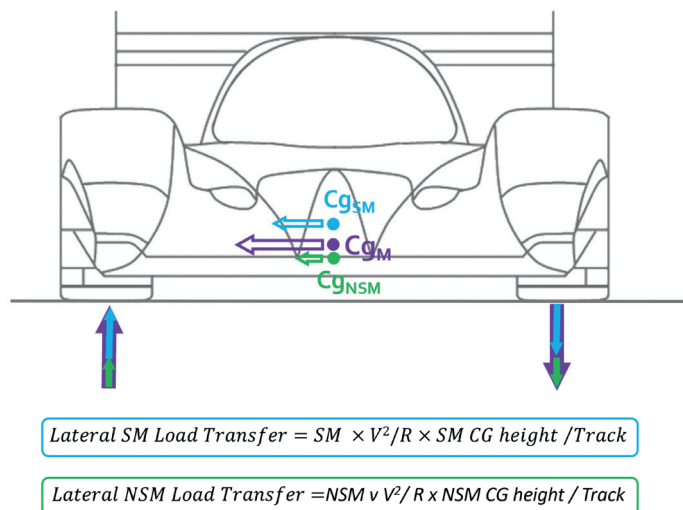
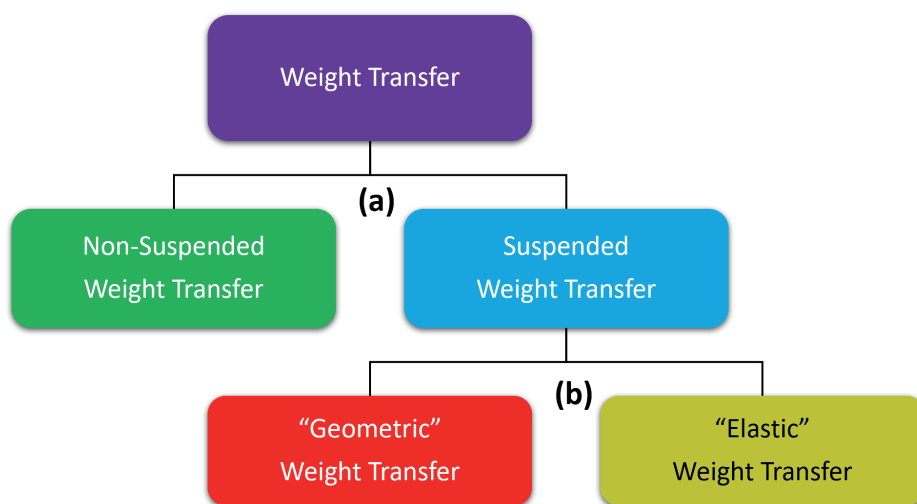


Fig2: Simplified 2D view equations of the decomposition of load transfer situations

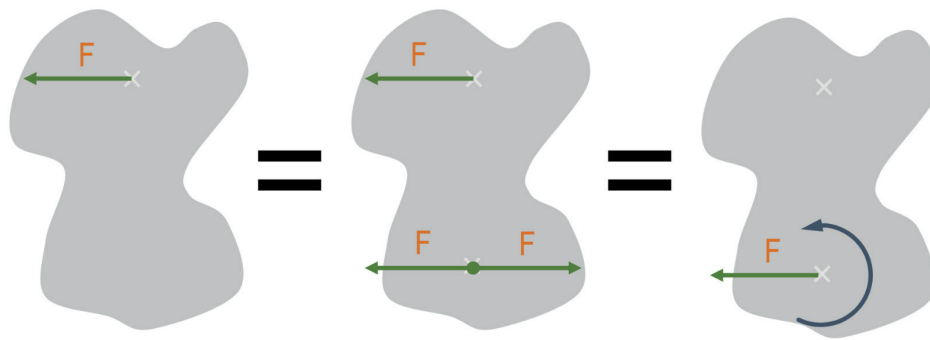


(a): decomposition of the total lateral load transfer in a non-suspended and a suspended mass load transfer

(b): decomposition of the suspended mass load transfer in a geometric load and an elastic load transfer

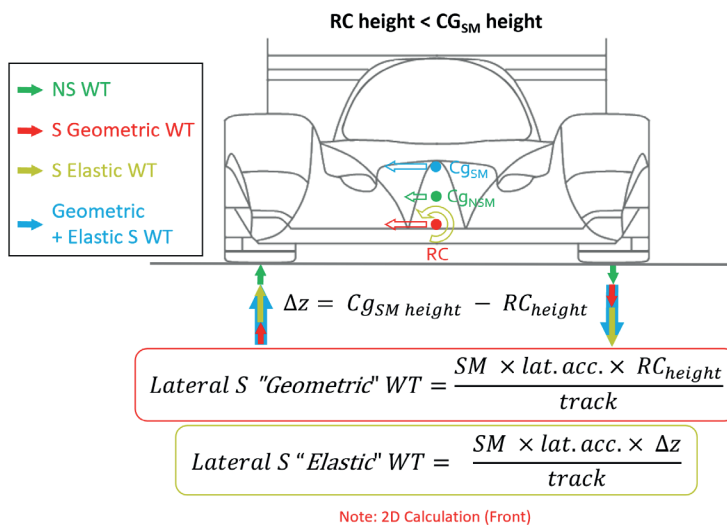
The higher the roll centre, the less roll moment and elastic roll transfer, but the more geometric load transfer

Fig 3: The basic principle of force



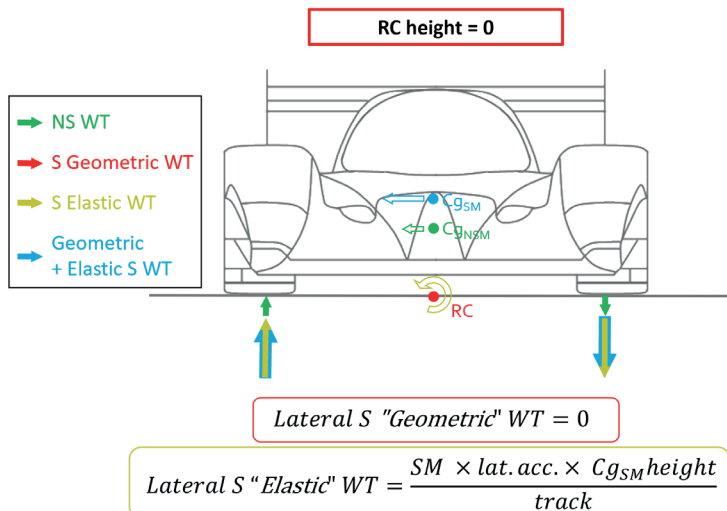
A force applied at one point of a body is equal to the same force applied at another point of that body and a torque

Fig 4: Simplified 2D equation of suspended mass load transfer in geometric and elastic load cases



Decomposition of the suspended mass load transfer in geometric (red), and in elastic (yellow) conditions

Fig 5: Simplified 2D equation of load transfer with the roll centre at ground height



In this condition, the suspended mass load transfer (blue) is elastic (yellow) and there is no geometric load transfer (red)

The fact is the kinematic roll centre is not a real, physical point of the body. But there are still a few useful perspectives to be shared

Once we take the tyres' lateral force, as well as the concepts of force-based roll centre and jacking forces (a topic for a future article), we will have a more exact appreciation of the load transfer principles.

But my experience is that for novices, this simplified explanation is a good, necessary (but insufficient) first step to better understand load transfers.

4. One of these simplifications is shown in **Figures 3 and 4**. These say that a force applied at one point of a body (the suspended mass c of g) is the same as the equal force applied at another point of the body (the kinematic roll centre) and a torque. The fact is the kinematic roll centre is not a real, physical point of the body. But, there are still a few useful perspectives to be shared.
5. If we look at the equations, and the vertical force vectors at each tyre base, we realise that if the roll centre is on the ground (**Figure 5**) the totality of the suspended mass load transfer is elastic. 100 per cent of suspended mass load transfer (blue) is in fact an elastic load transfer (yellow) and there isn't any geometric load transfer (red).
6. In the (very hypothetical) case of a roll centre at the same height as the suspended mass c of g (**Figure 6**), there isn't any roll moment because the distance between the suspended mass c of g and the roll centre is zero. There won't be any roll angle. Of course, there will still be a roll angle due to the tyres' deflection and the chassis and suspension elements compliance. But whether the spring and anti-roll bars are stiff or not, it does not make any difference. If there isn't any roll moment (action), there isn't any kinematic roll angle (reaction). And here is an important concept: if there isn't any roll moment and

So far we have been looking at the 'picture' (steady state), but once we look at the 'movie' (transient) things are quite different

any roll angle, there are no spring compression (outside) or extension (inside) nor any anti-roll bar twisting. A strain gauge on a pushrod would not record any force variation.

But how then is the load transferred from the inside to the outside wheel? Answer: through the wishbones.

8. That is why we speak about geometric load transfer. What determines the amount of geometric load transfer? The roll centre altitude. What determines the roll centre position? The geometry of the suspension.

Similarly, we speak about elastic load transfer because the roll moment will be reacted by the elastic component of the suspension: springs, anti-roll bars, maybe bump stops (if they are engaged) and, in transient, dampers.

9. If the roll centre was at half the distance between the ground and the suspended mass c of g , 50 per cent of the suspended load transfer would also be, and 50 per cent geometric, and the other 50 per cent would be elastic.
10. Last illustration of the distribution of geometric and elastic suspended mass load transfers the roll centre under the ground (Figure 7). In this case, the whole suspended mass load transfer (blue) is divided in a negative geometric (red) load transfer and a bigger, positive elastic load transfer (yellow). But the total remains the same, because the load transfer (at least in this simplified 2D, one-track example) is essentially depending on the mass, car speed and the corner radius, thus lateral acceleration ($A = V^2/R$), the c of g height and the track width.

So, why do we care about splitting the load transfer in a non-suspended and suspended load transfer, and then the suspended one in a geometric and an elastic load transfer, if at the end the total of these load transfers remains the same? Because they do not grow at the same rate, that's why. So far, we have been looking at the 'picture' (steady state), but once we look at the 'movie' (transient), things are quite different.

In Figure 8 we can see the evolution of the different parts of the suspended mass load transfer vs time by observing the variation of the left front tyre load in a right-hand corner under a simplified open loop input of an increasing, and then decreasing lateral acceleration (light brown) from 0.5 to 2.5 seconds with a peak at 2g.

Geometric is shown in red, and the three components of the elastic load transfer: anti-roll bar in blue, spring in green and damper in purple, with a roll centre above the ground.

Some interesting observations here:

1. The shape of this lateral acceleration is roughly what we see in a corner with a peak around the corner apex.
2. This is a simplified open loop. Simple then complicated, not the other way! In the real world, the lateral acceleration

depends, in a closed loop, on the tyre lateral forces, which in turn depend in part on the tyres' vertical load, which depend on the load transfers, which depend on the lateral acceleration.

3. The non-suspended load transfer is not displayed here.
4. We see the geometric load transfer (red) has the same shape as the lateral acceleration. It starts and ends at the same time. There is no delay or sluggishness, unless there would be

Figure 6: Simplified 2D equation of load transfer with the roll centre at the same height as the c of g

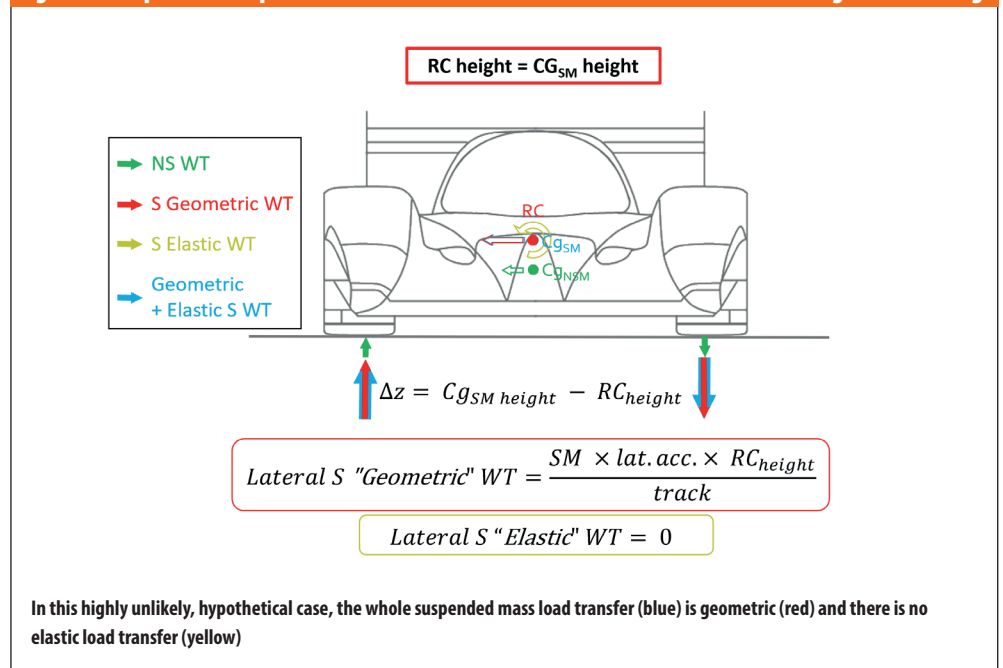
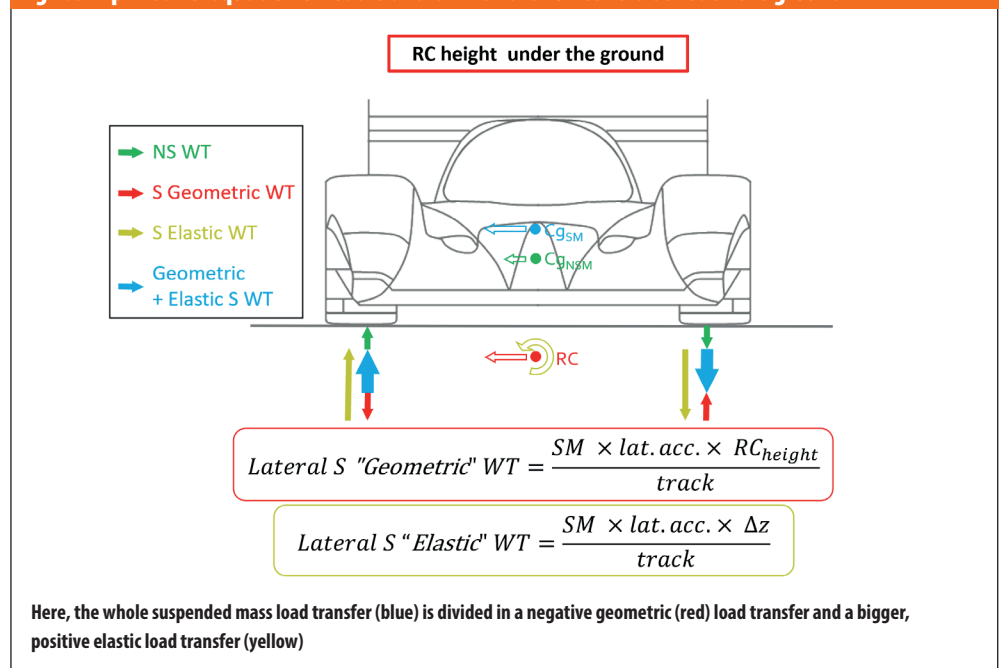


Fig 7: Simplified 2D equation of load transfer with the roll centre beneath the ground





The effect of the springs and ARB are biggest in the region of peak lateral acceleration (corner apex)

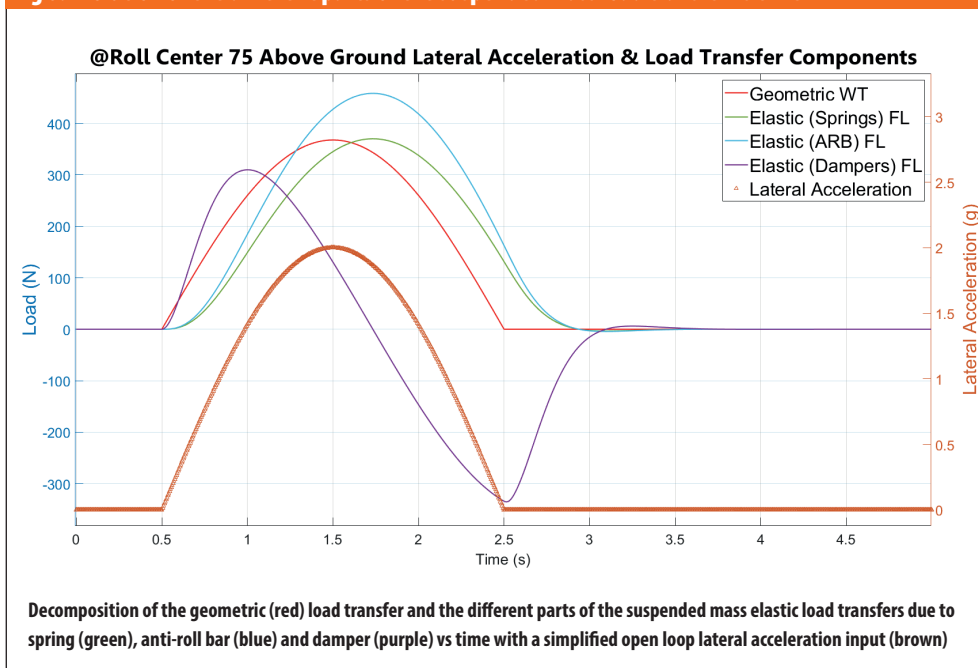
- some elasticity on the suspension elements, which is the case on a passenger car with suspension bushings.
5. If we look at the peaks, there is more load transfer from the ARB (blue) than the springs (green). That is usually the case in rear-wheel drive racecars.
 6. Suspended mass elastic load transfers that go through springs, ARB and the dampers do not stop when the lateral acceleration does. In fact, it takes about one second for the system to stabilise. That is all due to the suspended mass roll inertia.
 7. The peaks of the suspended mass elastic load transfers that go through the ARB (blue) and springs (green) occur later

- (about 0.25 second in this case) than the peak of the lateral acceleration. Again, due to the suspended mass inertia.
8. While the effect of the springs and ARB are biggest in the region of peak lateral acceleration (corner apex), they are negligible at corner entry and exit.
 9. However, that is the contrary for the dampers. At peak lateral acceleration (in fact, a touch later due to the suspended mass roll inertia), the roll angle is the biggest, but the roll speed changes sign and goes from zero. No roll speed, no damper speed, no damper effect.

10. If the effect of the dampers is close to negligible in the region of the peak lateral acceleration, it is considerable at corner entry and exit. In the first 0.5 second the dampers do, for better or worse, much more than the springs and the ARB.
11. If you pay attention to the first one or two tenths of a second, in the very first metres of corner entry, you'll see the geometric load transfer (red) does much more than the springs (green), the ARB (blue) and even the dampers (purple). We then realise how suspension kinematics will have a considerable effect on car behaviour.

We will go into detail on this next month. 

Fig 8: Evolution of the different parts of the suspended mass load transfer vs time



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